

CAIRO UNIVERSITY- FACULTY OF ENGINEERING- GENERAL- SYSTEMS & BIOMEDICAL ENGINEERING PROGRAM- SBE 2023-ALTERNATIVE (8 SEMS)

FRESHMAN FALL semester 1

	Course Code	Course Title	CHs	Contact Hours				OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs	
				Lecture	Tut (2)	(3)			Off. Tut								Off. Hrs
						App. Tut	Lab	Stud									
1	MTHG001	Algebra	2	1		3				4	111	4	40	20	40	2	
2	PHYG001	Mechanical properties of Matter, Waves and Thermodynamics	3	2		1.5	1.5		1	5	115	5	40	20	40	3	
3	EMCG001	Engineering Mechanics-Statics	2	1		3				4	98	4	40	20	40	2	
4	INTG005	Introduction to Computer Science	3	2			3			5	112	5	40	20	40	3	
5	MTHG002	Calculus 1	3	2	2					4	112	5	40	20	40	3	
6	INTG001	Technical Drawing	2	1		1.5	1.5			4	100	4	40	20	40	2	
7	GENG001	Critical & Creative Thinking	2	2						2	97	2	40	20	40	2	
			17	11	2	9	6	0	1	28	745	29					

FRESHMAN SPRING semester 2

	Course Code	Course Title	CHs	Contact Hours				OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs	
				Lecture	Tut (2)	(3)			Off. Tut								Off. Hrs
						App. Tut	Lab	Stud									
1	MTHG003	Calculus 2	3	2	2					4	106	5	40	20	40	3	
2	PHYG002	Electricity and Magnetism	3	2		1.5	1.5		1	5	115	5	40	20	40	3	
3	EMCG002	Engineering Mechanics- Dynamics	3	2		3				5	115	5	40	20	40	3	
4	CHEG001	Chemistry for Engineers	2	1				3		4	103	4	40	20	40	2	
5	ENGG001	Applied+ and Modern Manufacturing Engineering	3	2		3				5	115	5	40	20	40	3	
6	GENG002	Societal Issues	2	2						2	97	2	40	20	40	2	
7	ELEG102	Introduction to Computer Architecture	3	1	2				1	3	99	4	30	20	50	2	
			19	12	4	7.5	1.5	3	0	28	750	30					
			19	12		(9/12)				28							

SOPHOMORE FALL semester 3

	Course Code	Course Title	CHs	Contact Hours					OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)			Off. Tut	Off. Hrs							
						App. Tut	Lab	Stud									
1	MTHG102	Linear Algebra and Multivariable Integrals	3	2	2						4	110	5	40	20	40	3
2	ELEG101	Electrical Circuits	3	2		1.5	1.5				5	111	5	30	20	50	2
3	SBEG101	Medical Physics	3	2		1.5	1.5				5	111	5	30	20	50	2
4	SBEG103	Systems Physiology	3	2	2						4	110	5	30	20	50	2
5	SBEG105	Intro to Imaging and Image-based Anatomy	2	2							2	108	3	30	20	50	2
6	SBEG107	Programming Principles	3	2		1.5	1.5				5	111	5	30	20	50	2
7	GENG111	(UR3) Selected Topics 1	2	2							2	96	2	40	20	40	2
			19	14	4	4.5	4.5	0	0	0	27	757	30				

SOPHOMORE SPRING semester 4

	Course Code	Course Title	CHs	Contact Hours					OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)			Off. Tut	Off. Hrs							
						App. Tut	Lab	Stud									
1	MTHG103	Ordinary Differential Equations & Mathematical Transforms	2	1		3					4	108	4	40	20	40	2
2	ELEG103	Analog & Digital Electronics	3	2		1.5	1.5				5	111	5	30	20	50	2
3	EPEG130	Introduction to Electrical Power and Machines Engineering	3	2	2						4	108	4	30	20	50	2
4	SBEG102	Introduction to Medical Equipment	2	2							2	96	3	30	20	50	2
5	SBEG104	Medical Data Structure	3	2	2						4	108	4	30	20	50	2
6	SBEG106	Introduction to Biomechanics	3	2	2						4	108	5	30	20	50	2
7	SBEG108	Numerical Methods in BME	3	2		1.5	1.5				5	111	5	30	20	50	2
			19	13	6	6	3	0	0	0	28	750	30				

SOPHOMORE Summer semester

	Course Code	Course Title	CHs	Contact Hours					OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)			Off. Hrs	Off. Hrs							
						App. Tut	Lab	Stud									
1	SBEG281	Industrial Training - 1	1								0						
			1	0	0	0	0	0	0	0	0						

JUNIOR FALL semester 5

	Course Code	Course Title	CHs	Contact Hours				OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs	
				Lecture	Tut (2)	(3)			Off. Tut								Off. Hrs
						App. Tut	Lab	Stud									
1	MTHG202	Special Functions and Partial Diff Eqs	2	1		3				4	100	4	40	20	40	2	
2	SBEG201	Biotransport	3	2	2					4	111	4	30	20	50	2	
3	SBEG203	Bio-Sensors	3	2		1.5	1.5			5	111	5	30	20	50	2	
4	SBEG205	Biological Signal Processing	3	2		1.5	1.5			5	111	5	30	20	50	2	
5	SBEG207	Algorithms in Medical Applications	3	2	2					4	111	4	30	20	50	2	
6	SBEG209	Biostatistics	3	2	2					4	111	4	30	20	50	2	
7	GENG1xx	Restricted Elective-1	2	2						2	96	2	40	20	40	2	
			19	13	6	6	3	0	0	28	751	30					

JUNIOR SPRING semester 6

	Course Code	Course Title	CHs	Contact Hours				OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)										
						App. Tut	Lab	Stud	Off. Tut							
1	SBEG202	Clinical Engineering	3	2	2					4	111	4	30	20	50	2
2	SBEG204	Bio-Measurements	3	2	2					4	111	4	30	20	50	2
3	SBEG206	Biochemistry & Molecular Biology	2	2						2	98	3	30	20	50	2
4	SBEG3XX	DISCIPLINE elective-1	3	2	2					4	111	4	30	20	50	2
5	SBEG3XX	DISCIPLINE elective-2	3	2	2					4	111	4	30	20	50	2
6	SBEG3XX	DISCIPLINE elective-3	3	2	2					4	111	4	30	20	50	2
7	GENG2xx	Restricted Elective-2	2	2						2	97	3	40	20	40	2
			19	14	10	0	0	0	0	24	750	26				

JUNIOR Summer semester

	Course Code	Course Title	CHs	Contact Hours					OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)			Off. Tut	Off. Hrs							
						App. Tut	Lab	Stud									
1	SBEG282	Industrial Training 2	2								0						
			2	0	0		0	0	0	0	0						

SENIOR-1 FALL semester 7

	Course Code	Course Title	CHs	Contact Hours					OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)			Off. Tut	Off. Hrs							
						App. Tut	Lab	Stud									
1	SBEG3XX	DISCIPLINE elective-4	3	2	2						4	111	4	30	20	50	2
2	SBEG3XX	DISCIPLINE elective-5	3	2	2						4	111	4	30	20	50	2
3	SBEG4XX	PROGRAM elective-1	3	2	2						4	111	4	30	20	50	2
4	SBEG4XX	PROGRAM elective-2	3	2	2						4	111	4	30	20	50	2
5	SBEG4XX	PROGRAM elective-3	3	2	2						4	111	4	30	20	50	2
6	GENG2xx	Free Elective	2	2							2	95	3	40	20	40	2
7	SBEG481	Graduation Project 1(Biodesign: The Process of Innovating Medical Technologies)	2	2							2	98	3				
			19	14	10	0	0	0	0	0	24	748	26				

SENIOR-2 SPRING semester 8

	Course Code	Course Title	CHs	Contact Hours				OFFICE (1)		Total Contact Hrs	SWL	ECTS	Class work	Mid Term	Final	Exam Hrs
				Lecture	Tut (2)	(3)										
						App. Tut	Lab	Stud	Off. Tut							
1	SBEG4XX	PROGRAM elective-4	3	2	2					4	111	4	30	20	50	2
2	SBEG4XX	PROGRAM elective-5	3	2	2					4	111	4	30	20	50	2
3	SBEG4XX	PROGRAM elective-6	3	2	2					4	111	4	30	20	50	2
4	SBEG4XX	PROGRAM elective-7	3	2	2					4	111	4	30	20	50	2
5	SBEG4XX	PROGRAM elective-8	3	2	2					4	111	4	30	20	50	2
6	GENG351	Selected Current Local Issues	1	1						1	80	1	40	20	40	3
7	SBEG482	Graduation Project 2	3	1		6				7	120	9				
			19	12	10	6	0	0	0	28	755	30				

TOTAL

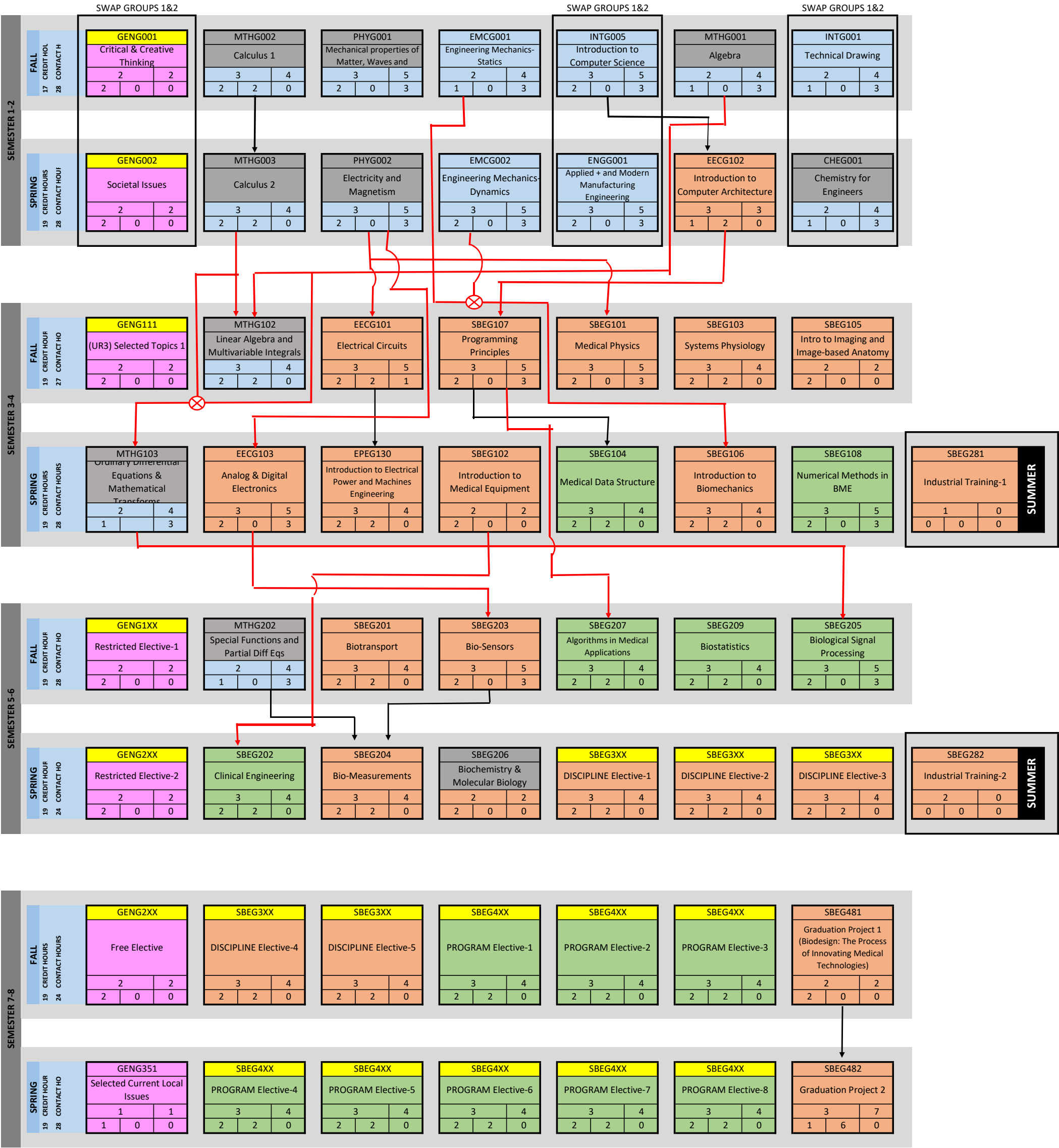
153

215

600

231

CAIRO UNIVERSITY- FACULTY OF ENGINEERING- GENERAL- SYSTEMS & BIOMEDICAL ENGINEERING PROGRAM- SBE COURSE MAP 2023-ALTERNATIVE (8 SEMESTERS)



SWL is calculated according to TOR 2022.

CREDITS	153
CONTACTS	215

ELECTIVES	45
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COURSE MAP LEGEND															
UNIVERSITY REQ. (U)	<table border="1"><tr><td colspan="3">COURSE CODE</td></tr><tr><td colspan="3">COURSE NAME</td></tr><tr><td colspan="2">CR</td><td>CNTC</td></tr><tr><td>LEC</td><td>TUT(2)</td><td>TUT/LAB/STU D(3)</td></tr></table>			COURSE CODE			COURSE NAME			CR		CNTC	LEC	TUT(2)	TUT/LAB/STU D(3)
COURSE CODE															
COURSE NAME															
CR		CNTC													
LEC	TUT(2)	TUT/LAB/STU D(3)													
FACULTY REQ. (FR)															
DISCIPLINE REQ. (DI)															
PROGRAM REQ. (PR)															
ELECTIVE															
BASIC SCIENCE (BS)															